

Task 4: Aggregate Functions & Grouping

Practical SQL Solutions & Data Analytics Concepts

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TOOLS USED
DB Browser for SQLite

KEY CONCEPTS
Grouping, Summarization,
Filtering

Part 1: SQL Query Implementations

1. Employee Metrics and Salary Averages by Department

```
-- Calculate total headcount, sum, and formatted average salary per department
SELECT
    dept_id,
    COUNT(*) AS total_staff,
    SUM(emp_salary) AS total_dept_budget,
    ROUND(AVG(emp_salary), 2) AS mean_salary
FROM employee_records
GROUP BY dept_id;
```

2. Filtering High-Earning Departments using HAVING

```
-- Identify departments where average earnings exceed 50,000 threshold
SELECT
    dept_id,
    ROUND(AVG(emp_salary), 2) AS avg_salary
FROM employee_records
GROUP BY dept_id
HAVING AVG(emp_salary) > 50000;
```

3. Department Maximum Salary Extraction

```
-- Determine highest individual salary in each operational department
SELECT
    dept_id,
    MAX(emp_salary) AS peak_salary
FROM employee_records
GROUP BY dept_id;
```

4. Counting Unique Designations Across Organization

```
-- Count total number of unique job titles stored in database
SELECT COUNT(DISTINCT designation) AS total_unique_roles
FROM employee_records;
```

Part 2: Interview Questions & Technical Explanations

1. What is GROUP BY?

The GROUP BY statement organizes rows that share identical values in specified columns into summary groups. It is predominantly used alongside aggregate functions like SUM () or COUNT () to calculate metrics per group.

2. Difference between WHERE and HAVING?

WHERE operates as a row-level filter before data is grouped, and cannot evaluate aggregate functions. In contrast, HAVING filters group-level results after aggregation takes place.

3. How does COUNT(*) differ from COUNT(column)?

COUNT (*) evaluates every single record in the dataset regardless of content, including NULL values. COUNT (column_name) specifically ignores NULL entries within that designated column.

4. Can you group by multiple columns?

Yes, specifying multiple columns in GROUP BY creates sub-groups for every distinct combination of those attributes (e.g., grouping by department_id and location).

5. What is ROUND() used for?

ROUND () is a mathematical function that truncates or rounds numeric figures (such as floating-point calculations) to a specified number of decimal places.

6. How do you find the highest salary by department?

Use the MAX () function grouped by the department column:

```
SELECT department_id, MAX(salary) FROM employees GROUP BY department_id;
```

7. What is the default behavior of GROUP BY?

It summarizes identical key values into single output rows. In standard relational SQL, it collapses duplicates without altering overall filtering rules unless paired with HAVING.

8. Explain AVG and SUM.

SUM () aggregates all values in a numeric column into a grand total. AVG () calculates the arithmetic mean by dividing the sum of non-null values by their count.

9. How to count distinct values?

Include the DISTINCT keyword inside the aggregate call, like COUNT (DISTINCT column_name) , to ensure duplicate entries are excluded from the count.

10. What is an aggregate function?

An aggregate function takes multiple input values from a column across several records and yields a single consolidated summary value (e.g., AVG, COUNT, MAX, MIN, SUM).